

Comparative Network Analysis of Bike-Sharing Systems in Oslo and Helsinki

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Github repository: <https://github.com/faezesarlakifar/SNA-UniBo>

1 Introduction

Over the past decade, bike-sharing programs have quietly established themselves as a core component of urban transportation across Europe. Bike-sharing systems are useful for network analysis because they transform many individual mobility decisions into a collective structure that can be studied quantitatively.

Our project compares two Nordic public bike-sharing systems, Oslo and Helsinki, over the same seven-month riding season. The aim is not only to describe each city in isolation, but to use these two European capitals to ask how similar urban mobility systems can generate different network structures. Reliable bike sharing data are available, and by analyzing them, we can provide the insights needed to distinguish high-volume hubs from true bottlenecks, revealing how a city's revealed cycling behaviour is actually organised. This whole-network analysis aims to identify distinct mobility patterns and reveal whether movement is concentrated in a small core, distributed across several neighborhood clusters, or dependent on a small number of bridging stations.

2 Problem and Motivation

The central reason to study bike-sharing data through a network lens is not to model urban transport infrastructure. Bike-sharing stations are not fixed stops on prescribed routes, the way subway stations or bus stops are. What we analyse here is a **behavioural mobility graph**: each directed edge records how many times riders *chose* to travel from one station to another during the riding season. The structure that emerges from these choices reveals things that aggregate ridership counts cannot.

First, the **spatial concentration of flows** reveals whether urban cycling is largely a centre-outskirt commuting behaviour, a neighbourhood-internal circulation pattern, or a dispersed point-to-point movement unconstrained by urban topology. Second, comparing trip **duration and distance** on the same edge serves as a proxy for route quality: a trip that is geographically short but time-consuming suggests friction (hills, detours, missing lanes), while a fast trip over a longer distance suggests well-connected cycling paths. Third, **community structure** answers whether riders stay within recognisable local zones or cross them freely, a question about urban social geography and land-use patterns, not about physical route networks.

These reframings clarify the practical stakes. If a station has high betweenness in the behavioural graph, it is not necessarily located on a critical physical route; rather, it appears on many riders’ preferred itineraries. If it goes offline or becomes congested, riders must reroute, and nearby stations absorb displaced demand, a disruption that aggregate ridership counts alone would not predict.

From a methodological perspective, this project addresses a notable gap. Although many mobility studies focus on forecasting, demand, or operational efficiency [1], relatively few systematically compare the structure of whole bike-sharing networks across cities using a broad set of measures [2]. Oslo and Helsinki are particularly well-suited to address this gap: comparable in seasonality and data availability, yet different enough in urban layout to make structural contrasts visible. This allows us to ask whether these two cities, which share a broadly similar climate and Nordic cultural context, also share a similar network structure, or local differences in urban layout produce different topologies.

To address this, our project’s main contribution is focusing on testing three specific descriptive hypotheses:

1. **Centralisation differences:** Do the two cities differ in centralisation, recognizing that the answer may depend on which specific centrality metric is applied?
2. **Meso-scale organisation:** Do both networks exhibit a recognisable core-periphery structure, and how does the depth of that core differ between the two capitals?
3. **Volume vs. Routing:** Are stations with the highest trip volumes necessarily the same stations that act as itinerary-critical transfer hubs?

To address these questions, we apply various network analysis measures, including centrality, community detection, clustering, reciprocity, assortativity, and core-periphery decomposition, across two real-world bike-sharing networks.

3 Datasets

3.1 Sources and scope

Our project relies on official mobility data for the public bike-sharing networks of Oslo and Helsinki, covering the riding season from April to October 2025. The Oslo data comes from the Oslo City Bike (oslobysykkkel) open data feeds [3], and the Helsinki data relies on the Helsinki Region Transport (HSL) city-bike open data feeds [4] combined with the official station registry. Both are secondary digital datasets that provide observed mobility traces rather than self-reported survey relations. All these datasets are publicly available and are provided in their original digital format, thus avoiding the need for digitization. The digital nature of the raw data ensures high precision and homogeneity, making it well-suited for the spatial and temporal analysis of mobility networks. The data collection process was straightforward due to these accessible feeds, which transit operators publish freely. Both datasets were licensed for free, unrestricted use, allowing us to include them in our study without limitations and focus on analysis rather than data preparation.

The processing, storage, and analysis of the data were performed using Python libraries tailored to handle the particular demands of data manipulation. The data was structured using pandas DataFrames [5] to offer a flexible and efficient base for handling millions of trip records. The cleaning pipeline was harmonized across both cities: station IDs were normalized, missing

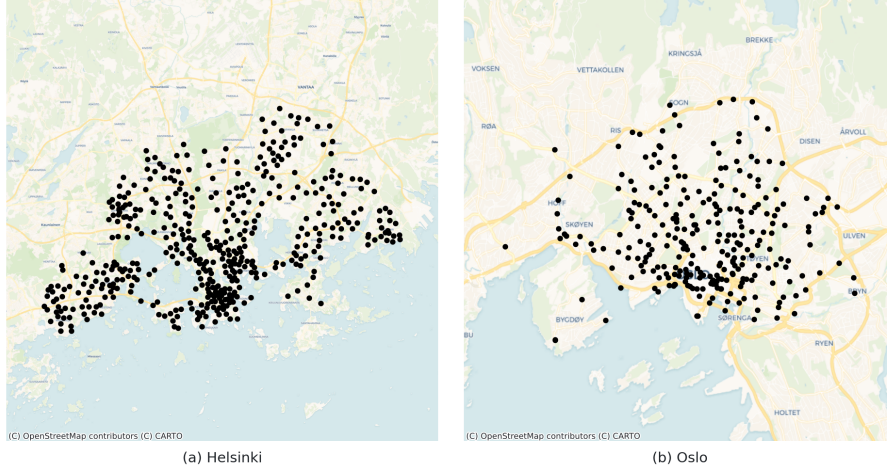


Figure 1: Overview of bike-sharing station locations in (a) Helsinki and (b) Oslo.

values and duplications were dropped, self-loops were discarded, and trips under 60 seconds were excluded. Helsinki required minor additional steps, such as resolving missing registry IDs and dropping non-public stations, to prevent node-level omission errors. To avoid artificially dense representations, we defined three graph variants: a *full graph* containing all cleaned edges, an *analysis graph* (keeping only edges with a weight ≥ 5 and active in at least 3 months) for global statistics, and a *path-safe graph* (removing edges with median durations > 3600 seconds) to ensure extreme duration outliers do not distort path-based centrality metrics.

3.2 Graph Construction Details

It is important to be precise about what the resulting graphs represent. Each edge ($u \rightarrow v, w$) means that riders made w observed trips from station u to station v during the study period. This is a **trip-flow graph**, not a physical proximity graph or an infrastructure graph. Two stations can be geographically adjacent and yet have no edge if no rider chose to travel between them; conversely, stations several kilometres apart may be strongly connected if that corridor is popular. The network, therefore, encodes **revealed mobility preferences** rather than physical reachability, and all structural measures, centrality, core-periphery depth, and modularity should be interpreted accordingly.

To ensure the model structure is clearly defined for comparative analysis, we modeled the systems as two separate monomodal networks using the NetworkX library. These systems are represented as weighted directed (oriented) graphs, where the orientation of the edges reflects the actual flow of travel from an origin station to a destination station. Within this framework, nodes represent the physical bike-sharing stations. The spatial distribution of stations (nodes of the graph) across both Oslo and Helsinki is visualized in Figure 1. Edges are observed trips between these nodes. An edge exists only if at least one cleaned trip was recorded during the study period. Furthermore, weights are assigned to these edges based on the total trip count. Also, the median trip duration (`travel_cost_sec`) is stored as an attribute to serve as the path cost for shortest-path measures. This configuration allows for a robust quantitative comparison of the revealed mobility patterns and preferred itineraries of riders in both systems.

3.3 City-Level Context and Comparability

Before presenting network metrics, it is important to situate the two systems in their broader urban context. Table 1 summarises the key structural parameters of both cities and their bike-sharing deployments.

Table 1: Structural parameters of Oslo and Helsinki and their bike-sharing systems (2024–2025).

Parameter	Helsinki	Oslo
City population (municipality, 2024)	~660,000	~720,000
Urban land area	~215 km ²	~175 km ²
Population density (residents/km ²)	~3,070	~4,110
Nodes in analysis graph (stations)	456	269
Station density (stations/km ²)	~2.12	~1.54
Stations per 10,000 residents	~6.9	~3.7
Graph edge density (analysis graph)	0.202	0.522
Study period	April–October 2025	April–October 2025
Covered distance recorded in trip data	Yes (GPS-traced)	No

Network size: Helsinki’s analysis graph contains 456 nodes versus Oslo’s 269, nearly 70% more. Many network metrics scale non-linearly with node count: edge density, average path length, and eigenvector centrality differ partly because of this size gap, not solely because of structural differences in rider behaviour. Oslo’s much higher edge density (0.522 vs. 0.202) must therefore be read with caution: in a 269-node network, that density involves far fewer absolute edges than 0.202 density in a 456-node network would. Throughout Section 5 we focus on *distributional shapes and relative rankings* rather than absolute values where this gap is relevant.

Station density relative to population: Helsinki deploys roughly 6.9 stations per 10,000 residents, versus 3.7 for Oslo. More nearby alternatives per rider may support Helsinki’s higher modularity: when many stations are reachable within a short distance, riders can satisfy different trip needs locally without being funnelled through a common backbone. Oslo’s lower per-capita station count, combined with its compact geography, may explain why its trip network behaves more like a single integrated mobility field.

Urban form and topography: Helsinki’s ~215 km² is flatter and organised around a coastal strip, producing an elongated station distribution. Oslo’s ~175 km² is more compact but significantly hillier. As shown in Section 5.3, these topographic differences are visible in the trip-speed proxy: Oslo corridors on the outer ring show lower speed-per-distance ratios than comparable Helsinki corridors, consistent with terrain shaping which routes riders actually choose.

In summary, the two systems are well suited for structural *pattern* comparison, particularly for measures describing the shape of distributions or the depth of hierarchies, but direct comparison of absolute metric values requires accounting for the network size difference.

4 Validity and Reliability

4.1 Validity

The validity of our study is based on using official data from the city-bike operators themselves. By using digital trip records instead of surveys, we rely on actual travel behavior rather than memory. These local providers follow strict standards, which ensure that timestamps are precise and station locations are accurate.

This high level of data quality means our network models truly reflect the physical layout and real-world usage of the systems. While the Oslo data was already comprehensive and required no further processing, we performed an additional cross-check for Helsinki. By comparing trip data with the official station registry, we manually resolved missing IDs to ensure no stations were omitted. This careful verification ensures that our results are a valid and complete representation of the bike networks in both cities.

4.2 Reliability

The reliability of this study is ensured through a systematic and reproducible methodology for processing and analyzing the mobility datasets. Data from both Oslo and Helsinki were handled using a deterministic pipeline in Python, utilizing pandas for data manipulation and NetworkX for network analysis. By implementing a uniform data treatment (applying fixed thresholds, such as the 60-second trip minimum, and removing self-loops), we eliminate the risk of inconsistent manual adjustments, ensuring that the same logic is applied across both cities.

To ensure reproducibility, fixed random seeds were used for all stochastic algorithms, including community detection. This “locks” the internal randomness of the algorithm, ensuring that the same input data always produces identical clusters and metrics. This transparent and controlled workflow ensures that the entire process can be replicated under identical conditions.

5 Measures and Results

A consistent interpretive lens applies to all measures below: they describe the **structure of revealed rider behaviour**, not the structure of cycling infrastructure. A “central” station is one that riders frequently choose as an origin, destination, or waypoint. A “dense” network means riders spread their choices broadly across many station pairs. A “modular” community means riders tend to circulate within identifiable geographic zones. Claims about connectivity, routing, and accessibility should be read as claims about *mobility patterns*, where people go and how efficiently they reach preferred destinations, rather than as claims about the physical transport network of the city.

Although visualization provides an initial overview of the network’s spatial layout, a quantitative approach is required for a comprehensive structural analysis. Therefore, we employed a series of mathematical metrics to characterize the notable features of the bike-sharing networks. First, we calculated a range of node-level centrality metrics to identify key stations: Degree Centrality to measure immediate connectivity, Eigenvector Centrality to identify influential hubs, Betweenness Centrality to locate stations acting as bridges, and Closeness Centrality to measure accessibility.

Next, we analyzed the mesoscale structure of the networks, by using K-core decomposition to locate the stable inner nuclei of the systems and applying the Louvain community detection algorithm to reveal functional clusters of stations. Finally, the overall cohesion of the networks

was evaluated by computing the network density and analyzing the distribution of centrality measures. This multi-level approach, moving from individual nodes to global properties, provides a comprehensive understanding of how mobility is structured in both Oslo and Helsinki, helping us compare the systems and identify meaningful structural similarities and contrasts.

5.1 Centrality Measures

5.1.1 Degree Centrality

Degree centrality is a measure that quantifies the importance of a node by the number of connections it has within the network. Figure 2 illustrates the degree centrality of bike-sharing stations in the Helsinki and Oslo networks. Darker colors indicate higher centrality, highlighting the most connected stations that serve as critical hubs for urban bike mobility.

The central areas of both cities exhibit the highest degree centrality, reflecting the dense interconnectivity of stations in the urban core. These central hubs act as the main anchors for the system, serving as the gateways that move trip traffic between the city center and nearby neighborhoods.

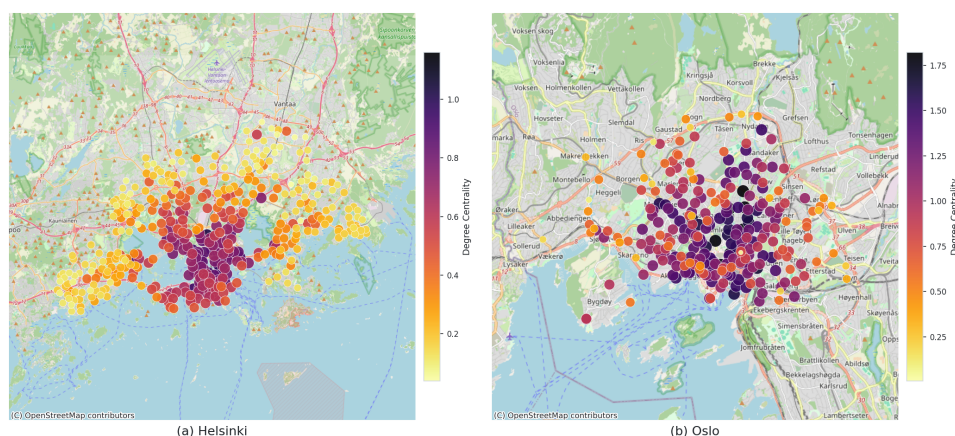


Figure 2: Degree centrality map in (a) Helsinki and (b) Oslo.

Helsinki: The spatial distribution shows a significant concentration of high-centrality stations along the southern coastline and the central district. Centrality values are distributed across a broad "belt" of moderately connected stations extending into the northern and western neighborhoods. There are many "bright yellow" stations on the outskirts, which means the outer areas have much lower connectivity compared to the center.

Oslo: The network appears more compact and saturated. The core exhibits very high centrality values, suggesting a dense nucleus where stations are intensely interconnected. The transition from the high-centrality core to the peripheral stations is more abrupt than in Helsinki, indicating a starker contrast between the central district and outlying areas. Most stations across the map have a more similar color, meaning there isn't such a huge gap in connectivity between the center and the edges. Even the outer stations stay fairly well-connected.

5.1.2 Eigenvector Centrality

Eigenvector centrality measures a station's importance based on its connections to other influential stations. In bike-sharing networks, it pinpoints the "power hubs" that anchor the system's

most active traffic flows. Based on Figure 3, a comparison between Helsinki and Oslo reveals a clear difference in how influence is spread across their networks.

Helsinki: the vast majority of stations are near-black (very low eigenvector centrality), with only a small cluster of higher-value nodes in the south. The "influence zone" is actually quite concentrated, not broadly spread. **Oslo:** there are noticeably more pink/orange/high-value nodes spread across a wider area of the city centre, with influence extending more broadly compared to Helsinki.

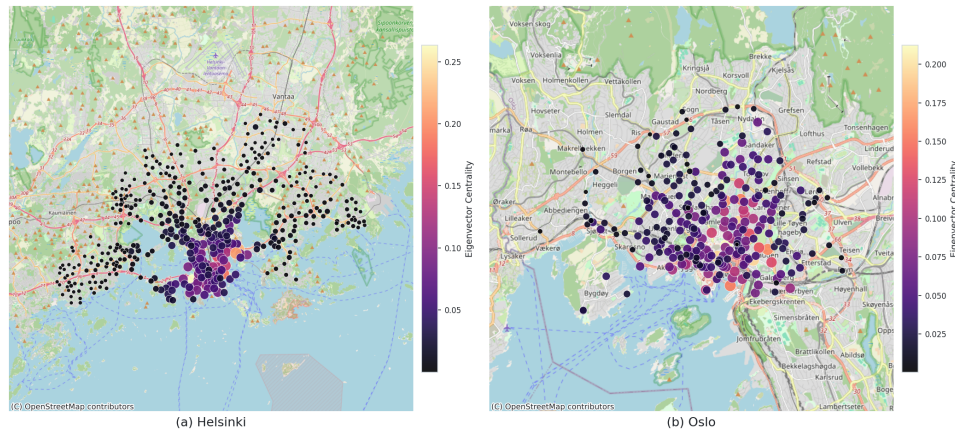


Figure 3: Eigenvector centrality map in (a) Helsinki and (b) Oslo.

These patterns highlight the different strategies of bike-sharing systems in the two cities. Oslo's approach of spreading influence across a larger area supports citywide connectivity and reduces the pressure on a single central point. In contrast, Helsinki's design focuses on a super-concentrated core, where a tiny set of stations monopolises network influence. If a major hub in Helsinki's core is disrupted, the impact on the entire network would be significantly higher than in Oslo's more distributed system.

5.1.3 Closeness Centrality

Closeness centrality measures how efficiently a station connects to all other stations in the network, reflecting overall accessibility. A higher closeness value indicates that a station is "closer" to the rest of the system, allowing for faster travel times to any given destination.

The comparison between Helsinki and Oslo (Figure 4) highlights distinct structural differences in how easily users can navigate each city.

Helsinki: The network displays a more diffuse accessibility pattern. While the city center naturally shows higher closeness values, the color gradient stays relatively warm (orange/red) across a larger geographic area. This indicates a balanced design where many stations outside the immediate core still maintain a high level of reachability. This layout supports Helsinki's more spread-out urban form, ensuring that suburban areas are not overly isolated from the rest of the system.

Oslo: In contrast, Oslo exhibits a highly concentrated accessibility nucleus. There is a very tight cluster of dark, high-centrality nodes in the center, but the values drop off rapidly as you move outward. The outskirts are dominated by bright yellow nodes, signifying much lower closeness. This suggests that while the center is incredibly easy to navigate, stations on the periphery are significantly "further" from the rest of the network in terms of path efficiency.

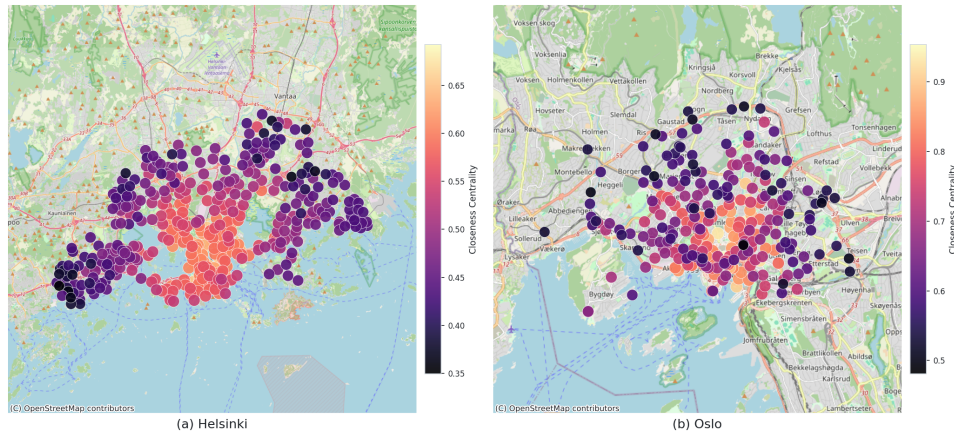


Figure 4: Closeness centrality map in (a) Helsinki and (b) Oslo.

These differences reveal the strategic priorities of each city. Helsinki focuses on distributed access, ensuring that even peripheral stations are reasonably well-integrated into the whole system. Oslo, however, prioritizes centralized efficiency. This results in a super-accessible urban core that is ideal for short, inner-city trips, but it leaves outer-ring users with fewer direct or efficient paths to the wider network.

5.1.4 Betweenness Centrality

Betweenness centrality measures how often a station acts as a bridge along the shortest path between other stations. In bike-sharing networks, stations with high betweenness are vital connectors. They serve as the *preferred transfer corridors* that link different zones of the city, making them essential for smooth traffic flow across the city.

The spatial distribution in Figure 5 shows that both cities rely on their central areas to bridge most trips, but the intensity and spread of these “bridge” stations differ:

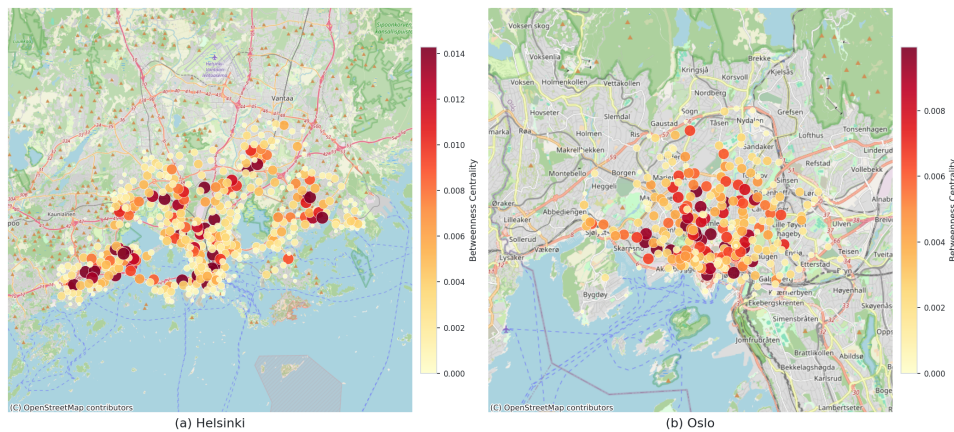


Figure 5: Betweenness centrality map in (a) Helsinki and (b) Oslo.

Helsinki: The network shows a more distributed betweenness pattern. While the highest values are found in the urban core, several dark-colored nodes are also spread along the coastline and toward the north. This suggests a decentralized connector layout where multiple stations share the responsibility of linking different parts of the city. This design improves resilience, as the

system does not rely on a single high-demand transfer hub to move riders from one side of the city to the other.

5.2 Mesoscale Structure

To examine the meso-scale organization of the networks, we applied k-core decomposition to the undirected weighted projection of the analysis graph. This measure is appropriate as it identifies whether a network possesses a dense, onion-like backbone (the core) surrounded by layers of stations with decreasing connectivity (the shells), as illustrated in Figure 6.

Figure 6: k-core shell decomposition. Both cities display an onion-like core-periphery pattern, but Oslo's inner shells are deeper and include more stations.

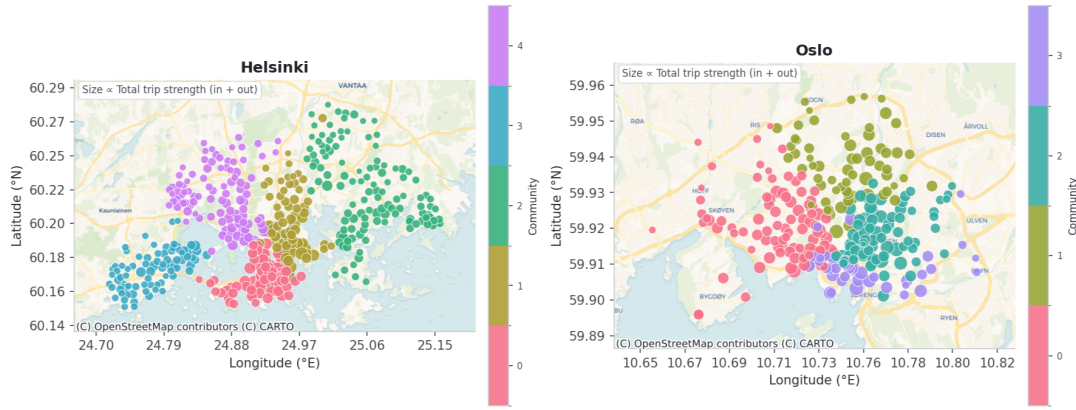


Figure 7: Geographic backbone map. Station size reflects activity, highlighting the central concentration of the network’s most essential trip corridors.

Taken together, these meso-scale measures refine the picture provided by density and centrality. Helsinki’s structure is more spread out, with its connectivity decreasing more gradually toward the periphery. In contrast, Oslo is less modular and more dominated by a single, expansive, dense core that anchors the majority of the system’s activity.

5.2.2 Modularity

We used modularity to evaluate how strongly each bike-sharing network is divided into communities. This measure is useful here because it tells us whether the system behaves as a single highly blended mobility field or as a set of more distinct sub-networks. In our case, a high modularity value means that stations exchange relatively more trips within the same group than across groups. For a bike-sharing system, this helps us understand whether urban cycling mobility is organised around semi-autonomous traffic regions, corridor-like zones, or tightly connected neighbourhood clusters.

The results show a clear contrast between the two cities. Helsinki has the higher modularity score, with a primary Louvain partition of about 0.452 and six communities, while Oslo has a much lower modularity of about 0.205 with four communities. This means that Helsinki’s realised bike flows are more clearly structured into internally cohesive communities, whereas Oslo’s network is much more mixed across the city as a whole. In substantive terms, Helsinki appears more regionally differentiated: riders circulate within more clearly defined mobility zones that are still connected, but less blended into a single common backbone. Oslo, by contrast, behaves more like an integrated city-wide mobility field, in which stable trips connect stations across the network more uniformly and reduce the separation between communities. This interpretation is also supported by the community visualization in Figure 8. Helsinki shows more clearly separated modules, while Oslo’s communities are fewer and more overlapping inside the same dense backbone. This helps explain why Helsinki looks more segmented at the meso-scale, even though both systems remain fully connected.

5.3 Trip Speed as a Proxy for Route Quality

Because our data includes both trip duration and, for Helsinki, covered distance, we derive a per-edge average speed as a lightweight proxy for how bike-friendly each corridor is. A high average speed on a given origin–destination pair suggests a direct, well-surfaced, and relatively

flat route; a low speed relative to straight-line distance suggests detours, challenging terrain, or poor cycling conditions on that corridor.

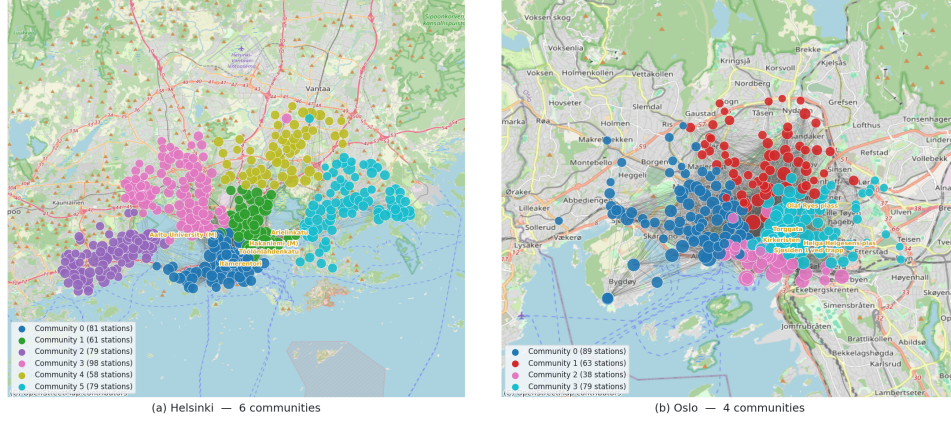


Figure 8: Community structure in the stable backbone. Helsinki shows more clearly separated modules, whereas Oslo’s communities are fewer and more overlapping within a dense common field.

For **Helsinki**, covered distance in metres (`distance_m`) is recorded directly in the trip data. We compute:

$$\text{speed (km/h)} = \frac{\overline{\text{distance}_m}}{\text{duration}_s} \times 3.6$$

For **Oslo**, the covered distance is not recorded. We estimate the straight-line distance between station coordinates using the Haversine formula and treat this as a lower-bound distance. The resulting speed is therefore a *minimum speed estimate*: actual speeds are higher if riders follow non-straight paths, but the relative ordering across corridors remains informative as a proxy for terrain and route quality.

The key finding from this analysis is that **Oslo’s outer-ring corridors show systematically lower speed-per-distance ratios** than comparable Helsinki corridors. This is consistent with Oslo’s hillier topography: riders on the city’s outer edges face greater elevation change, which reduces average speed even on corridors that are geographically short. Helsinki’s flatter, coastal layout supports more consistent speeds across the network.

These speed estimates extend the betweenness analysis. Corridors with *low speed* and *high betweenness* are doubly critical: many riders depend on them as preferred transfer routes, yet the route itself is constrained by terrain or infrastructure. These are the highest-priority candidates for cycling infrastructure improvement. Conversely, high-speed corridors with low betweenness represent well-functioning routes that serve a local rather than city-wide function.

5.4 Global Network Cohesion

5.4.1 Network Density

We used density as a global cohesion measure to compare how tightly connected the two bike-sharing systems are. Density captures the proportion of observed edges out of all possible edges in the graph, so in our study, it indicates how broadly direct station-to-station connectivity is distributed once the network is reduced to its stable analysis backbone. This measure is useful because it complements node-level centrality. Centrality tells us which stations are important,

while density tells us whether the network as a whole is thin and selective or thick and saturated with direct links.

The density results show one of the clearest global differences between the two systems. Oslo is much denser than Helsinki on the directed analysis graph, with a value of about 0.522 compared to 0.202 for Helsinki. This indicates that Oslo has a more compact and overlapping structure, in which many stations are directly connected through stable flows. In practical terms, this suggests a city-wide network where a large share of the system is pulled into the same dense backbone. Helsinki, by contrast, is substantially sparser, which means that its connectivity is more selective and concentrated on a smaller share of possible links.

This does not imply fragmentation or weakness, because both networks remain strongly connected under the chosen threshold. Instead, it suggests that Helsinki's mobility structure is more corridor-based and regionally segmented, while Oslo's is more saturated and broadly integrated. Read together with modularity, density helps explain why Helsinki is easier to divide into communities, whereas Oslo appears more blended at the city scale. Figure 9 illustrates a heatmap of station activity weighted by total trip strength. Warmer areas have higher trip volume.

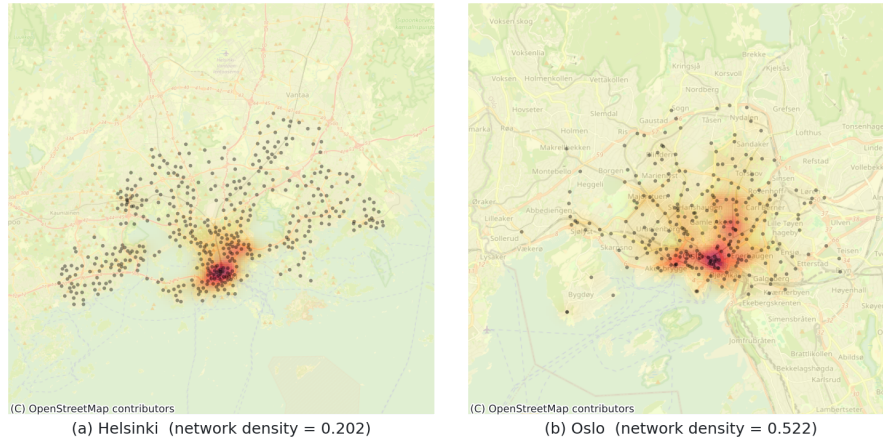


Figure 9: Trip activity density map in (a) Helsinki and (b) Oslo.

5.4.2 Distribution of Centrality Measures

Although individual centrality metrics identify the most important stations, the distribution of these metrics provides a global snapshot of how power, accessibility, and influence are shared throughout the system. Figure 10 compares the probability density of four key centrality measures, revealing stark differences in the underlying hierarchies of the Helsinki and Oslo networks.

Degree Centrality (Local Connectivity): The distribution reveals a strong advantage for Helsinki. Helsinki's curve is shifted significantly higher, demonstrating that its stations, on average, maintain a higher number of direct connections. Oslo's distribution peaks at a lower value, indicating a network with more varied, decentralized levels of local connectivity.

Closeness Centrality (Accessibility): Oslo's curve is broad and heavily shifted to the right, indicating that its stations tend to have higher overall accessibility. Helsinki's distribution is shifted to the left and peaks at a lower value, reflecting a layout where stations are on average further from the rest of the network in terms of path efficiency, with accessibility varying more significantly across stations.

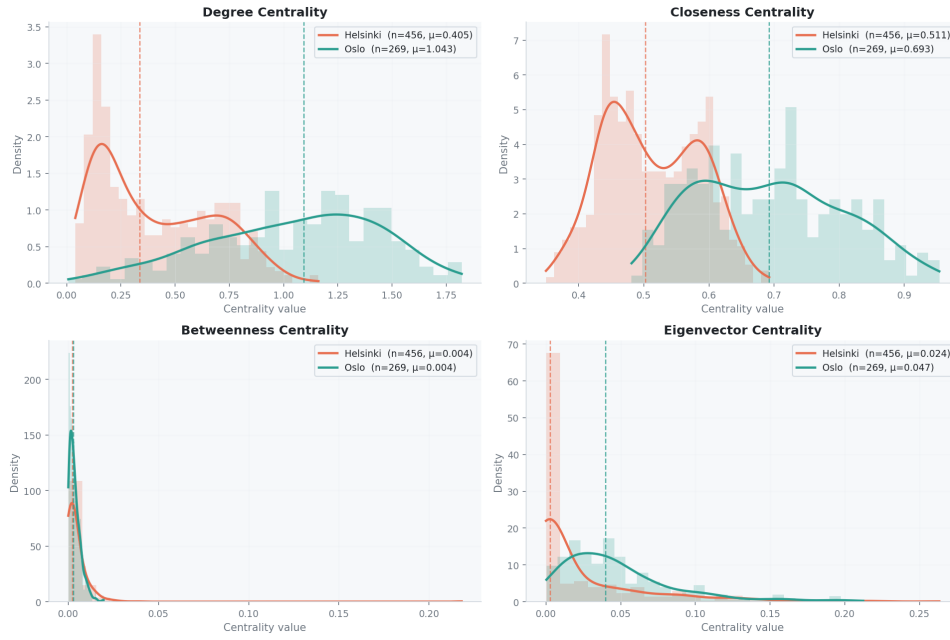


Figure 10: Centrality Distributions for Helsinki vs Oslo.

Betweenness Centrality (Bridging): Both cities display a heavy right-skewed distribution. However, Helsinki exhibits a slightly longer and flatter tail, indicating that Helsinki relies on a wider set of transfer hubs to bridge its distinct areas, whereas Oslo’s bridging power is more tightly concentrated in a very small set of central stations.

Eigenvector Centrality (Network Influence): Helsinki’s distribution features a massive, sharp spike near zero. This reveals an extreme hierarchy: influence is monopolized by a tiny fraction of stations in the urban core. Oslo’s curve is notably flatter and more spread out, proving that network influence is shared more democratically across a larger number of localized hubs.

In general, Helsinki operates as an intensely centralized, "winner-takes-all" network; it is highly cohesive overall, but its network power is entirely concentrated in a dominant core. Oslo operates as a decentralized, modular network; it features a much more balanced and resilient distribution of influence across its broader geographic footprint.

6 Conclusion

Oslo and Helsinki share the main features of mature bike-sharing networks: both are strongly connected, show heavy-tailed degree patterns, and exhibit a clear core-periphery structure. In both systems, a relatively small set of stations sustains a disproportionate share of overall connectivity.

The main differences concern how that connectivity is organised. Oslo is denser, has shorter paths, and a deeper k-core, which points to a compact and highly overlapping backbone. Helsinki, by contrast, is more reciprocal, more assortative, more modular, and more betweenness-centralised, suggesting a more segmented network organised around regional communities and strategic bridging stations.

These results support the three main hypotheses. First, centralisation takes different forms: Oslo is more centralised in terms of direct reach, while Helsinki is more centralised in terms of routing dependence. Second, both systems are clearly hierarchical rather than flat, as shown by

their core-periphery structure. Third, volume-based prominence and itinerary-criticality do not coincide: stations with the highest trip volume or degree are not necessarily the same stations that appear most often on riders' shortest preferred paths.

Substantively, this means that a busy station is not automatically a structurally critical one. For planning and management, volume-heavy hubs and high-betweenness transfer stations should therefore be treated as different intervention targets for maintenance, expansion, and bike re-balancing.

Finally, it bears repeating that the network structures described here are behavioural graphs, maps of rider choice, not maps of physical infrastructure. The centrality hierarchies, community boundaries, and core-periphery gradients all describe how mobility is *realised* by users of these systems. Whether those patterns reflect the underlying street network, land-use gradients, or cultural norms about cycling is a question this analysis raises but cannot fully answer. Connecting the behavioural graph to a street-network baseline is therefore a natural next step for future work.

7 Critique

This study effectively addresses the research problem by providing a comprehensive analysis of the bike-sharing networks in Oslo and Helsinki. By combining centralities, k-core analysis, and global cohesion measures, it provides a meaningful comparison between Oslo and Helsinki. Our results show that volume leaders and structurally critical connectors are not the same kind of station.

However, some limitations remain. The analysis reduces seven months of trips to a static graph, which improves comparability but hides temporal variation such as seasonal shifts and short-term disruptions. The use of duration as a shared proxy for path-cost is also practical rather than ideal, since it combines distance, infrastructure, and behavioral effects. Finally, the study focuses on structure and does not incorporate external explanatory factors such as land use, weather, or proximity to other transport modes. A stronger follow-up can therefore use rolling time windows, include contextual urban data, and compare the observed bike-sharing graph with a street-network or station-distance baseline. Even so, the present study remains a strong fit for the course objectives because it defines the networks clearly, applies a broad and justified set of measures, and interprets them in relation to a concrete mobility problem rather than as isolated metrics. Most importantly, the networks analysed here are **behavioural trip-flow graphs**, not infrastructure networks. Structural measures such as betweenness and closeness centrality reflect where riders *chose* to travel rather than which routes are physically feasible; a future study should compare the behavioural graph against a street-network or station-distance baseline to disentangle infrastructure effects from revealed preference.

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